IEEE Access
Multidisciplinary ⁝ Rapid Review ⁝ Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.XXXX.XXXXXXX

A Reproducible Retrospective Curtailment-Risk Benchmark and Fair Evaluation of GRU Learned-Space Retrieval on RTS-GMLC

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Funding and sponsor information must be confirmed by the authors before submission.

ABSTRACT Public curtailment-forecasting evaluations remain fragmented across private data, task-specific targets, and incomparable protocols. We build a retrospective benchmark from the first 8760 delivery-row-indexed observations in hashed 8784-row RTS-GMLC source files. A fixed 70% system-non-synchronous-penetration-type acceptance rule defines a method-independent proxy, and a transition slice identifies onsets. Source files lack forecast issue timestamps and data-vintage identifiers, so the 1 h and 24 h tasks are retrospective lags, not operational forecasts. The fair run separates fit, head-first checkpoint/hyperparameter selection, threshold calibration, and test with horizon-offset target boundaries across ten common GRU seeds. At the primary cap, mean-absolute-error (MAE) selection chooses retrieval-only (head weight zero) for every seed. Against the matched GRU head, retrieval lowers MAE at 1 h (mean paired difference -0.00496069) and 24 h (-0.00220055), with Holm-adjusted exact sign-flip p = 0.01171875 at each lag. This component result does not establish overall superiority: Persistence remains lower-MAE than selected GRU-LSR at both lags (0.00690794 versus 0.00777391 at 1 h; 0.02054651 versus 0.02076857 at 24 h). Onset-targeted selection is not estimable because selection and calibration contain zero positive onsets at both lags; the selected condition consequently equals the head (p = 1). A fixed 0.5 blend raises onset F1 at 1 h under that fallback but is null at 24 h, so neither comparison validates onset-targeted retrieval. Cap-level crossings are descriptive within the same 8760-row sequence. The benchmark supports a paired MAE mechanism result while preserving horizon- and metric-specific negative findings and a non-operational scope.

INDEX TERMS renewable energy curtailment, benchmark, reproducibility, analogue retrieval, time-series forecasting, transition detection, RTS-GMLC, naive baselines

I. INTRODUCTION

High instantaneous wind and solar shares can force curtailment when acceptance limits bind. International reviews document volumes and causes across multiple systems [1], national studies identify their drivers [2], and planning analyses show that some curtailment can be economically rational [3]. One explicit mechanism is the system non-synchronous penetration (SNSP) limit used on the all-island Irish system for frequency stability [4]. When the cap binds, curtailment follows from load and renewable trajectories, making cap-driven risk forecastable. Day-ahead SNSP forecasting has accordingly been demonstrated on Irish system data [5].

Within the curtailment-forecasting studies cited here, evaluations use task-specific signals, datasets, and protocols; the reviewed corpus does not document a released benchmark that combines an onset slice with seeded inference. The onset slice is sparse in the held-out primary-cap data: 57 of 2627 targets at 1 h and 172 of 2604 targets at 24 h. Forecasting-methodology research shows why strong naive baselines and temporally separated tests matter on sparse, persistent series [7], [8]. RTS-GMLC supplies time-synchronized day-ahead scenario series for load, wind, and solar [6]. The completed run uses the first 8760 of 8784 aligned source rows, whose recorded delivery keys end on December 30; it is not a complete calendar-year sample. The inspected assets also do not preserve forecast issue times or vintages.

A second, older gap concerns how forecasting models should use historical analogues. Retrieving similar past situations is a recurring idea in power-system forecasting and decision support. Its uses range from similar-day load forecasting in early expert systems and case-based reasoning for network operations [10] to analog ensembles in weather and renewable forecasting [9] and learned similarity on power signals. The cited studies validate retrieval on study-specific tasks and horizons. Within this bounded corpus, we did not identify a checkpoint-matched comparison of the same retrieval path at 1 h and 24 h. This is a search-bounded positioning statement, not an exhaustive state-of-the-art exclusion.

This paper evaluates retrieval as a conditional mechanism rather than assuming that it is uniformly useful. Three research questions organize the study:

- **RQ1:** Under a symmetric temporal gate and MAE-based selection, does learned-space retrieval improve continuous proxy MAE relative to its matched GRU head at 1 h and 24 h?
- **RQ2:** Does the available pre-test support permit a valid onset-targeted retrieval estimate, and what metric-specific results remain when the declared fallback is triggered?
- **RQ3:** Do the descriptive GRU-LSR-versus-Persistence orderings remain unchanged when the acceptance cap is varied on the same fixed 8760-row sequence?

The contributions follow those questions:

1. **A method-agnostic retrospective benchmark with an explicit information boundary.** The fixed proxy target is separated from model fitting. Fit, selection, calibration, and test target sets are disjoint, and horizon offsets place each downstream query endpoint within its phase. A target-hour direct transform is retained as a privileged visibility audit, not a forecaster.
2. **A matched retrieval-control design.** GRU-LSR uses a shared frozen GRU encoder, a fit-only $k = 8$ target bank, and selected or fixed head weights. For each objective, the GRU-head checkpoint is selected first; retrieval weights and matched controls then share that checkpoint, and inference is paired by training seed.
3. **An evidence-ranked result that keeps failures in scope.** The fair run supports lower MAE for retrieval relative to the matched head at both lags, but Persistence remains lower-MAE at the primary cap. Zero pretest onset support makes onset-targeted selection inapplicable, and cap crossings remain descriptive within one fixed source sequence.

The method claim is deliberately incremental. GRU-LSR introduces no new GRU cell, similarity loss, or neighbor rule; the contribution is the auditable benchmark and checkpoint-matched mechanism evaluation. Because raw-feature and randomized-encoder retrieval controls were not rerun under the fair gate, the experiment identifies the evaluated retrieval prediction path relative to its head, not a causal advantage of learned-space geometry over alternative retrieval spaces.

Section II reviews the relevant literature. Sections III–V define the benchmark, model, and fair comparison. Section VI answers the research questions in evidential order, and Sections VII–IX interpret the findings, limitations, and conclusion.

II. RELATED WORK

A. CURTAILMENT FORECASTING AND RENEWABLE ACCOMMODATION ASSESSMENT

The empirical curtailment literature is predominantly retrospective. Bird et al. [1] review international levels, causes, and mitigation, while Luo et al. [2] analyze China's Three-North wind curtailment. Yasuda et al. [11] relate annual curtailment rate to renewable share, Frew et al. [3] study economically optimal marginal curtailment, and Newbery [12] examines Irish wind build-out against interconnection and storage. These studies quantify or explain curtailment rather than predict its short-horizon onset.

The predictive strand in the cited corpus is system-specific. O'Sullivan et al. [4] establish the frequency-stability basis of the Irish SNSP limit — an operational constraint that motivates cap-based risk. Cardo-Miota et al. [5] forecast day-ahead SNSP trajectories for the same system with a machine-learning pipeline; its continuous penetration target makes it a close comparator in this corpus, but it does not use the onset protocol studied here. Hadian and Naderkhani [13] compare classical and deep models for curtailment time-series point forecasting on CAISO data and report GRU as the lowest-error model among their evaluated conditions. Related net-load forecasting work [14] anticipates the accommodation problem but does not map forecasts to the transition target defined here. The benchmark contribution is therefore positioned against this cited set; it does not claim an exhaustive literature search, state-of-the-art exclusion, or external expert validation.

B. RETRIEVAL, ANALOGUES, AND METRIC LEARNING IN POWER SYSTEM DECISION SUPPORT

Reasoning from similar past operating states is a recurring proposal with a long pedigree. Rahman and Bhatnagar [15] encoded similar-day operator heuristics in a 1988 expert-system load forecaster; Mandal et al. [16] retrieved Euclidean similar days to drive several-hour-ahead neural load forecasting; Lora et al. [17] forecast next-day market prices purely by weighted nearest neighbors over historical price trajectories. Case-based reasoning carried the idea into operations proper: Xu et al. [10] run the full retrieve–reuse–revise–retain cycle over network operating cases for coordinated voltage control. In weather-driven forecasting, the analog ensemble of Delle Monache et al. [9] retrieves the most similar past forecasts and uses their verifying observations as a predictive distribution, with transfers to wind power [18] and solar power [19]. Other work learns similarity through Siamese embeddings for appliance identification [20] and power-quality disturbance classification with k-NN [21]. GRU-LSR does not use a Siamese, contrastive, or pairwise objective; its space is learned only through the forecasting loss.

Across the reviewed retrieval studies, validation is typically confined to one task and one horizon or narrow horizon band on study-specific data. Matched non-retrieval controls are also uncommon, leaving the contribution of retrieval itself difficult to isolate. This paper therefore compares retrieval-only, equal-blend, and head-only predictions at 1 h and 24 h under a shared temporal gate. It also reports when the data cannot support an objective-matched onset comparison.

C. NAIVE BASELINES AND BENCHMARK DESIGN IN TIME-SERIES FORECASTING

Forecasting competitions motivate the benchmark protocol. M3 shows that sophisticated methods do not necessarily beat simple references [22]. Hyndman and Koehler demonstrate that common errors can degenerate on intermittent, near-zero series and propose scaled evaluation [23]. Later M-competitions compare entrants with naive combinations at scale [7], [24], [25].

In energy forecasting, GEFCom2014 establishes competition-grade protocol design [26], and community guidance requires verification against persistence and other standardized references [27]. Surveys identify temporal leakage, missing naive baselines, and inappropriate metrics as recurrent failures [28]. Kapoor and Narayanan document similar reproducibility failures across machine-learning science [8]. RTS-GMLC [6] supplies the public, synchronized load and renewable data needed for an auditable benchmark.

Recent IEEE Access load-forecasting studies continue to combine temporal representation learning with tensor graph convolution or TimesNet–Crossformer–LSTM stacks [29], [30]. Their prediction targets differ from curtailment onset, but they reinforce a relevant evaluation requirement: learned methods should be tested against naive and linear references. The fair experiment here answers a narrower mechanism question by holding the GRU checkpoint fixed across retrieval controls and retaining Persistence, Seasonal-24h, and Ridge as external references. Broader architecture comparisons remain in the historical supplement and are not mixed into the fair statistical family.

This literature motivates three design choices: persistence and seasonal-naive are first-class methods; onset metrics complement aggregate error on the sparse target; and learned-method comparisons use disjoint temporal phases, common seeds, and multiplicity correction. It also motivates the conservative interpretation in Section VI: a component can improve on its matched head without beating the naive reference, and an onset metric cannot rescue an objective arm that has no positive pre-test examples.

D. GAP STATEMENT

At the intersection, two gaps emerge from the bounded literature set reviewed here. **(G1)** The cited studies do not document a released curtailment-risk benchmark that combines a transition protocol with seeded statistics. **(G2)** Within the cited retrieval studies, we did not identify a multiple-lag public-benchmark evaluation with checkpoint-matched retrieval controls and an explicit record of pre-test event support. Contribution 1 addresses G1 within a retrospective scope, while Contributions 2–3 address G2 through the implemented comparison and its negative onset-support finding. Because no systematic-review protocol or independent domain-expert audit was executed, G1 and G2 are corpus-bounded positioning claims rather than proof of global novelty.

III. THE CURTAILMENT-RISK BENCHMARK

A. METHOD-INDEPENDENT TASK CONSTRUCTION

The benchmark uses the first 8760 aligned delivery rows from hashed RTS-GMLC files that each contain 8784 source rows [6]. The manifest records delivery keys from January 1 through December 30, so the executed subset is not described as a complete calendar year. System-aggregated load, wind, and solar define a fixed curtailment-risk proxy rather than observed dispatch curtailment. For renewable availability r_t, load d_t, and acceptance cap c, the accepted amount and proxy rate are

$$u_t = \min(r_t, cd_t), \qquad y_t = \frac{\max(0, r_t - u_t)}{\max(1, r_t)}.$$

The primary cap is $c = 0.70$; $c \in \{0.60, 0.80\}$ is used only for descriptive sensitivity. The cap is a benchmark policy parameter motivated by SNSP-class operating limits [4], not a universal physical limit. The target is computed once before fitting and is shared by every condition.

Each query contains 48 delivery rows of seven features: load, wind, PV, nonnegative net load, net-load ramp, renewable share, and an engineered branch-informed stress feature. Let $n_t = \max(0, d_t - r_t)$, let $R = \sum_j R_j$ be the sum of positive branch contingency ratings, and let $\bar{o} = \sum_j R_j(o_j^{\mathrm{perm}} + o_j^{\mathrm{tran}})/R$. The seventh feature is

$$z_t = \min\left(1,\ 0.55\frac{d_t}{R} + 0.25\bar{o} + 0.20\frac{|n_t - n_{t-1}|}{\max(1, d_t)}\right).$$

This is an engineered input constructed from static branch aggregates and delivery-row load dynamics; it is not a power-flow, contingency, or feasibility calculation. A window ending at row s predicts the target at delivery row $t = s + h$, where $h \in \{1, 24\}$. The source rows contain calendar delivery keys but no forecast-issue timestamp, as-of mapping, release identifier, or vintage. Accordingly, s is a benchmark index and h is a retrospective lag. Neither task is evidence of an operational forecast issued for delivery at t.

B. DISJOINT TEMPORAL GATE

The completed fair run places delivery-row boundaries at 4380, 5256, and 6132. Fit targets are below 4380; selection targets start at $4380 + h$; calibration targets start at $5256 + h$;

and test targets start at $6132+h$. These horizon offsets ensure that a query endpoint $s=t-h$ has reached the downstream phase boundary. They do not require all 48 historical rows in a query window to lie in the downstream phase; earlier-phase history within that window is permitted. Feature normalization, model fitting, and the retrieval bank use fit delivery rows only. Ridge penalties and GRU-head checkpoints use selection rows only; with the selected checkpoint fixed, retrieval head weights use the same selection rows. Detection thresholds use calibration rows only. Test rows are scored once after all artifacts are frozen (Fig. 1).

C. ONSET DEFINITION AND DECLARED FALLBACK

An onset is a transition from a quiet benchmark index to a material proxy at delivery:

$$\mathrm{onset}_t(h) \iff y_t \geq 0.02 \wedge y_{t-h} < 0.02.$$

Onset MAE is the mean absolute error restricted to these rows. Onset F1 thresholds continuous predictions into a transition classification. The frozen procedure selects a threshold from 40 prediction quantiles on calibration rows. If calibration contains no positive onset, it returns the fixed 0.02 fallback and records `fallback_no_positive_onsets`; it does not call that value calibrated. The onset-F1 selection objective uses the same rule on selection rows. When all candidate F1 values tie at zero, the declared ordering retains the first checkpoint (epoch 5) and first head-weight candidate ($\alpha=1$). This behavior is an auditable fallback, not evidence that the selected head is optimal for onset prediction.

The target-hour `DirectPolicyTransform-Privileged` condition applies the proxy equation to target-hour load, wind, and PV. It must have zero continuous error by construction. It is retained to expose the information available in the files, not as a lag forecaster. Its onset F1 can remain below one because the inherited classifier marks all threshold exceedances, including ongoing high-curtailment rows, whereas the onset target also requires the prior quiet-state condition.

D. DESCRIPTIVE CAP SENSITIVITY

The complete fair subset is rerun at caps 0.60, 0.70, and 0.80 on the same RTS-GMLC system and fixed 8760-row sequence. These are method-level scope checks, not independent system or year replications. No cross-cap p-values are computed, and no cap is selected after comparing test performance.

IV. GRU LEARNED-SPACE RETRIEVAL AND MATCHED CONTROLS

Figure 2 separates the method-independent target from the fitted retrieval mechanism. Let $X_s \in \mathbb{R}^{48\times 7}$ be the standardized query window and E_θ the GRU encoder. The direct head and fit-only retrieval estimate are

$$\hat{y}^{\mathrm{head}}_{s+h} = w_h^\top E_\theta(X_s) + b_h,$$

$$\hat{y}^{\mathrm{ret}}_{s+h} = \frac{1}{k}\sum_{q\in\mathcal{N}_k(s)} y_{q+h}, \qquad k=8,$$

where $\mathcal{N}_k(s)$ contains the nearest fit-bank embeddings under Euclidean distance. Query and bank windows use the same forecasting-trained, frozen encoder; no contrastive or pairwise loss is used. The evaluated prediction is

$$\hat{y}_{s+h}(\alpha) = \alpha\hat{y}^{\mathrm{head}}_{s+h} + (1-\alpha)\hat{y}^{\mathrm{ret}}_{s+h}.$$

The paper-facing name is **GRU learned-space retrieval (GRU-LSR)**. The historical string `DSTAR-GRU` is retained only in legacy supplementary archives. The fair run evaluates the conditions in Table 2. For each objective and seed, the checkpoint is first selected using GRU-head predictions on selection rows. That checkpoint is then frozen while the head-weight grid is evaluated, and all blend controls share it. Head-versus-retrieval contrasts therefore do not confound checkpoint choice, but they remain conditional on a head-selected checkpoint rather than a jointly retrieval-optimized checkpoint. Fixed $\alpha = 0$, 0.5, and 1 isolate retrieval-only, equal blending, and head-only prediction. Persistence, Seasonal-24h, and Ridge are comparison baselines. The privileged direct transform is not ranked as a forecaster.

The fair subset does not rerun the complete legacy v6 roster. It therefore cannot identify a new overall winner among the historical 14 methods or renew the old ablation claims. Those earlier results remain supplementary historical evidence and are not mixed into the fair-run statistical family.

This boundary also applies to mechanism attribution. The matched controls vary the prediction path after freezing a head-selected checkpoint; they do not compare learned-space retrieval with raw-feature k-NN, a randomized encoder, alternative distances, or a separately tuned k. Consequently, the experiment evaluates the complete GRU-LSR procedure but cannot decompose its retrieval gain into representation geometry, neighbor averaging, or regularization by fit-bank targets.

V. EXPERIMENTAL SETUP

A. COMPARISON BUDGET AND FAIRNESS

The GRU has one layer, hidden size 48, a linear head, a 20-epoch ceiling, and checkpoints at epochs 5, 10, 15, and 20. Ten common seeds are used. Training minimizes MSE with the frozen Adam equations, learning rate 10^{-3}, batch size 256, and predictions clipped to $[0,1]$. The retrieval bank is fit-only and uses $k = 8$. The head-weight grid is $\{1, 0.8, 0.6, 0.5, 0.4, 0.2, 0\}$. Ridge searches seven frozen penalties from 10^{-6} to 10. Both objective arms receive the same data phases and candidate budgets. The frozen sequence is head-checkpoint selection followed by head-weight selection, not joint checkpoint/weight optimization.

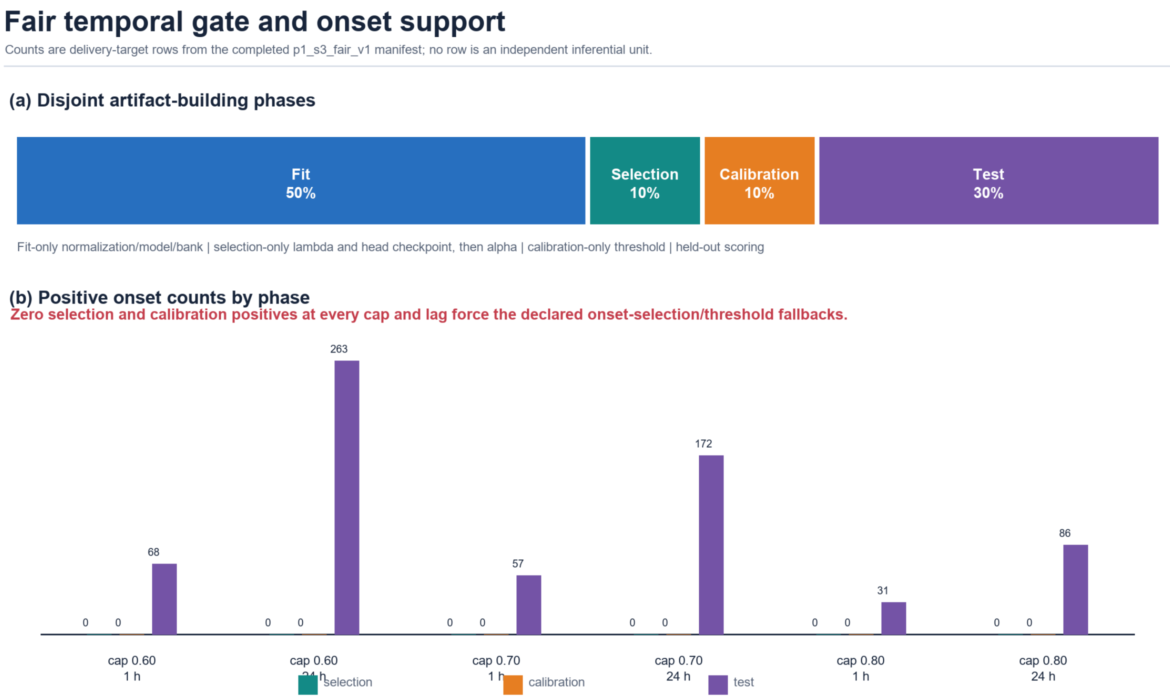

FIGURE 1. Manifest-derived phase design and onset support. Panel (a) shows the frozen phase budget. Panel (b) shows that selection and calibration contain zero positive onsets at every cap and lag, despite positive onset rows in the held-out test phase. Counts describe delivery targets; they are not independent inferential units.

TABLE 1. Phase and onset counts from `fair_onset_support.csv` at the primary cap.

Lag	Fit targets	Selection targets	Calibration targets	Test targets	Selection onsets	Calibration onsets	Test onsets
1 h	4332	875	875	2627	0	0	57
24 h	4309	852	852	2604	0	0	172

B. ESTIMAND AND ANALYSIS UNIT

The primary estimand is the mean paired within-seed treatment-minus-control difference, conditional on the fixed 8760-row sequence, cap, data phases, feature/metric definitions, head-first checkpoint-selection budget, retrieval bank, and lag. The analysis unit is one paired method-seed run. The 2627 and 2604 test targets at 1 h and 24 h are reused across seeds and are not treated as independent replicates. The two-sided sign-flip calculation enumerates all 2^{10} sign assignments of the ten paired differences. Its inferential interpretation assumes that, under a no-effect null for training randomness, the paired differences are sign-exchangeable across the frozen seeds; the seeds were common algorithmic initializations, not randomized experimental assignments. Holm adjustment covers the frozen six-contrast family separately within each lag. Lower differences favor treatment for MAE; higher differences favor treatment for onset F1. The mean paired difference is the scale-specific effect estimate. No confidence interval over seeds or data-resampling interval is available; the exact p-values and win/tie counts characterize only the ten frozen training seeds.

Persistence, Seasonal-24h, the target-selected Ridge conditions, and the direct transform have one deterministic row per cap and lag. Their comparisons are descriptive and receive no seed-based p-value. Cross-cap comparisons are also descriptive. Seed uncertainty covers training randomness only; it does not cover hours, onset events, years, systems, vintages, or deployments.

C. REPRODUCIBILITY BOUNDARY

The completed manifest records 510 result rows and hashes the frozen inputs, script, configuration, and output tables. A separate execution rerun with the same script, configuration, source hashes, and seeds reproduced all non-timing fields in the 510 rows and produced byte-identical leaderboard, paired-inference, cap-sensitivity, and policy-audit tables. This supports computational reproduction of the scientific outputs; it is not an external investigator replication and does not broaden the data or operational scope. Environment versions, implementation timing, hashes, incident logs, and legacy v5/v6 history are confined to the supplementary audit.

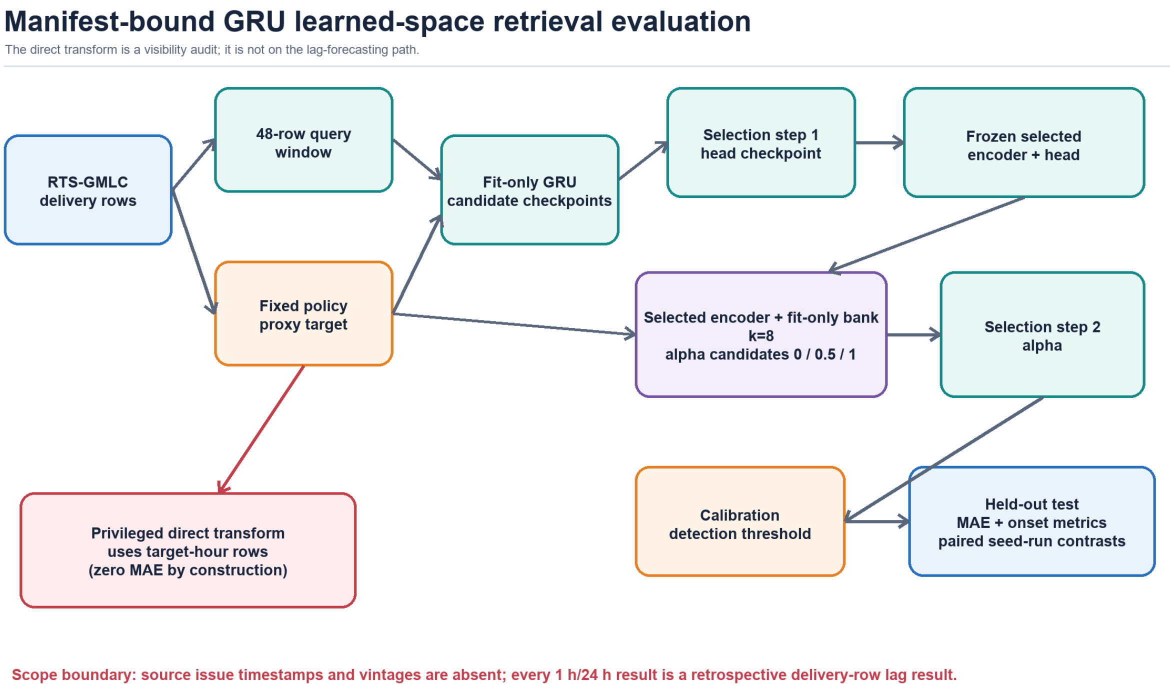

FIGURE 2. Fair-run architecture and information flow. The target-hour direct transform is a privileged visibility audit outside the lag-forecasting path. Selection and calibration are distinct, and held-out inference is paired by training seed.

TABLE 2. Fair-run conditions from `config.json` and `run_results.csv`.

Condition	Role	Frozen information use
GRU-LSR selected	Proposed condition	GRU-head checkpoint selected first, then α; fit-only learned-space bank
GRU-LSR Fixed0	Mechanism control	Retrieval only ($\alpha = 0$), same selected checkpoint
GRU-LSR Fixed0.5	Mechanism control	Equal head/retrieval blend, same selected checkpoint
GRU-LSR Fixed1	Mechanism control	GRU head only ($\alpha = 1$), same selected checkpoint
Ridge	Baseline	Fit-only coefficients; penalty selected on selection rows
Persistence	Baseline	Proxy at benchmark index s
Seasonal-24h	Baseline	Proxy at delivery row $t - 24$; equals Persistence when $h = 24$
Direct policy transform	Privileged control	Target-hour rows; zero continuous error by construction

VI. RESULTS

A. RQ1: DOES LEARNED-SPACE RETRIEVAL IMPROVE CONTINUOUS PROXY MAE RELATIVE TO THE MATCHED HEAD?

The privileged direct transform has zero MAE at both lags because it consumes the target-hour rows used by the proxy equation. That result verifies the construction path and simultaneously shows why it is not a lag forecaster. Among non-privileged conditions, Persistence has the lowest primary-cap MAE at both lags (Table 4 and Fig. 3).

MAE-based selection chooses $\alpha = 0$ in all ten seeds at both lags. The selected condition is therefore retrieval-only, not a successful head/retrieval mixture. Its mean MAE is 0.00777391 at 1 h and 0.02076857 at 24 h. Persistence remains lower at 0.00690794 and 0.02054651, respectively. The 24 h gap is small but adverse; no inferential claim attaches to the deterministic comparison.

The paired mechanism contrast nevertheless favors retrieval relative to the matched head in all ten seeds at both lags (Fig. 4). The mean selected-minus-head difference is -0.00496069 at 1 h and -0.00220055 at 24 h; both have exact $p = 0.001953125$ and Holm-adjusted $p = 0.01171875$. Fixed 0.5 blending also improves on the head in all pairs (-0.00269047 and -0.00122621), while retrieval-only improves on fixed 0.5 in all pairs (-0.00227022 and -0.00097435); the same adjusted p-value applies to each MAE contrast. RQ1

TABLE 3. Frozen fair-run hyperparameters and statistical contract.

Item	Value
Series / input	8760 delivery rows; 48-row windows; seven features
Retrospective lags	1 h and 24 h
Caps	0.70 primary; 0.60 and 0.80 descriptive sensitivity
Temporal phases	boundaries at 50% / 60% / 70%; downstream query endpoint offset by h; earlier history in a 48-row window permitted
GRU	one layer; hidden 48; epochs 20; checkpoints 5/10/15/20
Training	MSE; Adam; learning rate 10^{-3}; batch 256; ten common seeds
Retrieval	fit-only bank; Euclidean $k = 8$; head-first checkpoint selection; selected and fixed $\alpha \in \{0, 0.5, 1\}$ controls
Selection objectives	MAE and onset F1
Primary analysis	six paired GRU contrasts per lag at cap 0.70
Test / multiplicity	two-sided exact sign-flip; Holm within each lag

TABLE 4. Primary-cap continuous-error readouts from `fair_primary_cap_summary.csv`. Standard deviations are across ten training seeds where applicable.

Lag	Condition	Seeds	Head weight	MAE	SD
1 h	Direct transform (privileged)	1	–	0.00000000	–
1 h	Persistence	1	–	0.00690794	–
1 h	**GRU-LSR selected**	10	**0.0**	**0.00777391**	0.00016515
1 h	GRU-LSR fixed 0.5	10	0.5	0.01004413	0.00074902
1 h	GRU head	10	1.0	0.01273460	0.00152028
1 h	Ridge (MAE-selected)	1	–	0.01768352	–
1 h	Seasonal-24h	1	–	0.02036662	–
24 h	Direct transform (privileged)	1	–	0.00000000	–
24 h	Persistence / Seasonal-24h	1 each	–	0.02054651	–
24 h	**GRU-LSR selected**	10	**0.0**	**0.02076857**	0.00023138
24 h	GRU-LSR fixed 0.5	10	0.5	0.02174292	0.00058973
24 h	Ridge (MAE-selected)	1	–	0.02261682	–
24 h	GRU head	10	1.0	0.02296912	0.00117318

is therefore answered conditionally: retrieval improves the matched GRU head under MAE selection at both retrospective lags, but it does not beat Persistence at the primary cap and does not demonstrate blend synergy.

B. RQ2: CAN THE FAIR RUN ESTIMATE ONSET-TARGETED RETRIEVAL UTILITY?

No. Both selection and calibration contain zero positive onsets at 1 h and 24 h; the same is true at caps 0.60 and 0.80 (Fig. 1). The onset-F1 objective therefore ties across all checkpoint and blend candidates, retains epoch 5 and $\alpha = 1$ by frozen ordering, and uses the 0.02 threshold fallback. The selected onset condition is exactly the GRU head in every cap-0.70 pair. Its selected-minus-head onset-F1 difference is 0 at both lags (ten ties, Holm-adjusted $p = 1$). This is inapplicability evidence, not proof that retrieval has no onset effect.

The fixed controls remain useful diagnostics but do not repair the selection failure. At 1 h, fixed 0.5 blending raises test onset F1 over the head by 0.02441871 in all ten pairs (adjusted $p = 0.01171875$); because selected GRU-LSR equals the head, selected-minus-fixed-0.5 is -0.02441871 with the same adjusted p-value. At 24 h, fixed 0.5 minus head is +0.00163439 with five wins and five losses (adjusted $p = 1$), and the reverse selected-minus-fixed comparison is likewise null. Thus the horizon-specific result is positive only for a fixed 1 h diagnostic blend and null at 24 h, under an onset arm that had no positive selection or calibration examples.

The metric definition supplies a second negative result. The direct transform has exact continuous predictions yet onset F1 is 0.3333 at 1 h and 0.7527 at 24 h. The classifier flags ongoing above-threshold rows as positives, whereas the onset target additionally requires a quiet row at $t-h$. Consequently, onset F1 below one for this privileged condition is a metric-definition limitation, not direct-transform prediction error.

C. RQ3: DO GRU-LSR-VERSUS-PERSISTENCE ORDERINGS PERSIST ACROSS CAPS?

No stable ordering appears across the six cap-by-lag cells (Table 6 and Fig. 5). MAE-selected GRU-LSR is slightly lower-MAE than Persistence only at cap 0.60/1 h (-0.00010283) and cap 0.80/24 h (-0.00013934). Persistence is lower in the remaining four cells, including both primary-cap lags. These same-series crossings show cap sensitivity; they do not establish transport to another policy, year, or system.

The evidence hierarchy is therefore unambiguous. The strongest result is the paired within-seed MAE improvement over the matched GRU head. Persistence remains the lower-MAE primary-cap reference. Onset-targeted selection is unsupported, with a 1 h diagnostic fixed-blend improvement and a 24 h null result retained under that qualification. Cap crossings are descriptive only.

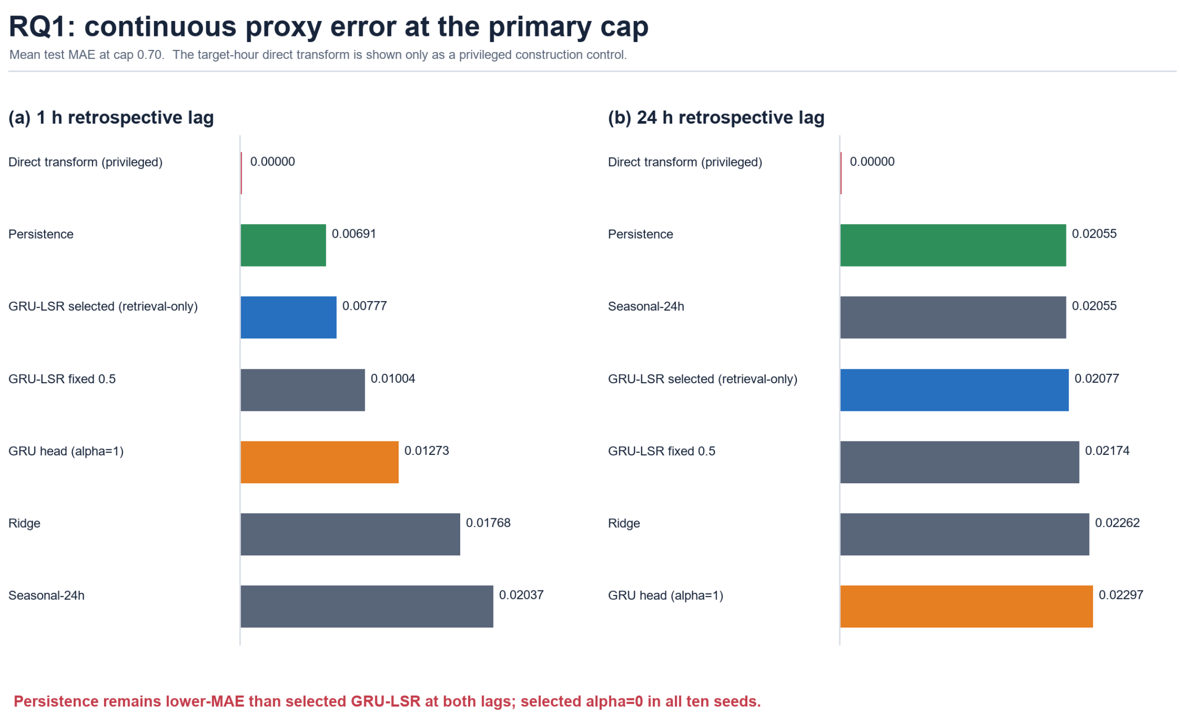

FIGURE 3. **Mean test MAE at cap 0.70. The selected GRU-LSR condition is retrieval-only in every seed. The direct transform is displayed as a privileged construction control and is not included in the forecasting comparison.**

TABLE 5. **Onset-F1 paired diagnostics at cap 0.70. Positive differences favor treatment, but the entire table is qualified by zero selection/calibration onset support.**

Lag	Treatment minus control	Mean difference	Treatment wins / ties / control wins	Holm p	Interpretation
1 h	Selected minus head	0.00000000	0 / 10 / 0	1.00000000	fallback identity
1 h	Fixed 0.5 minus head	+0.02441871	10 / 0 / 0	0.01171875	diagnostic improvement only
1 h	Selected minus fixed 0.5	-0.02441871	0 / 0 / 10	0.01171875	selected fallback is worse
24 h	Selected minus head	0.00000000	0 / 10 / 0	1.00000000	fallback identity
24 h	Fixed 0.5 minus head	+0.00163439	5 / 0 / 5	1.00000000	null
24 h	Selected minus fixed 0.5	-0.00163439	5 / 0 / 5	1.00000000	null

VII. DISCUSSION

A. WHAT THE PAIRED MAE RESULT SUPPORTS

Retrieval-only prediction improves on the matched GRU head in every seed at both retrospective lags. Because each comparison shares the head-selected checkpoint, the result isolates the prediction path more cleanly than a comparison between independently tuned models. It supports the proposition that averaging fit-bank targets from forecasting-trained embeddings can reduce the head's continuous proxy error on this fixed sequence, conditional on the head-first selection design.

The selected head weight of zero narrows the interpretation. The evidence favors the retrieval estimator, not a synergistic mixture of retrieval and the parametric head. Fixed 0.5 blending is better than the head but worse than retrieval-only in all MAE pairs. Claims that the blend combines complementary strengths would therefore exceed the executed result.

The same evidence does not establish that the *learned space* is the cause of the improvement. A parsimonious alternative is that fit-bank target averaging supplies nonparametric smoothing relative to the linear head, independent of whether the forecasting-trained embedding is uniquely useful. A fair raw-feature k-NN or randomized-space control would be required to separate those explanations. The present contribution is therefore an evaluation of GRU-LSR as implemented, not a new representation-learning theory or a state-of-the-art retrieval claim.

B. WHY THE ONSET QUESTION REMAINS UNANSWERED

Zero positive onset rows in both selection and calibration prevent objective-matched checkpoint selection, blend selection, and threshold calibration. The resulting head selection and 0.02 threshold are deterministic fallbacks. Test-set differences under those fallbacks describe what the frozen candidates did, but they cannot answer whether an onset-

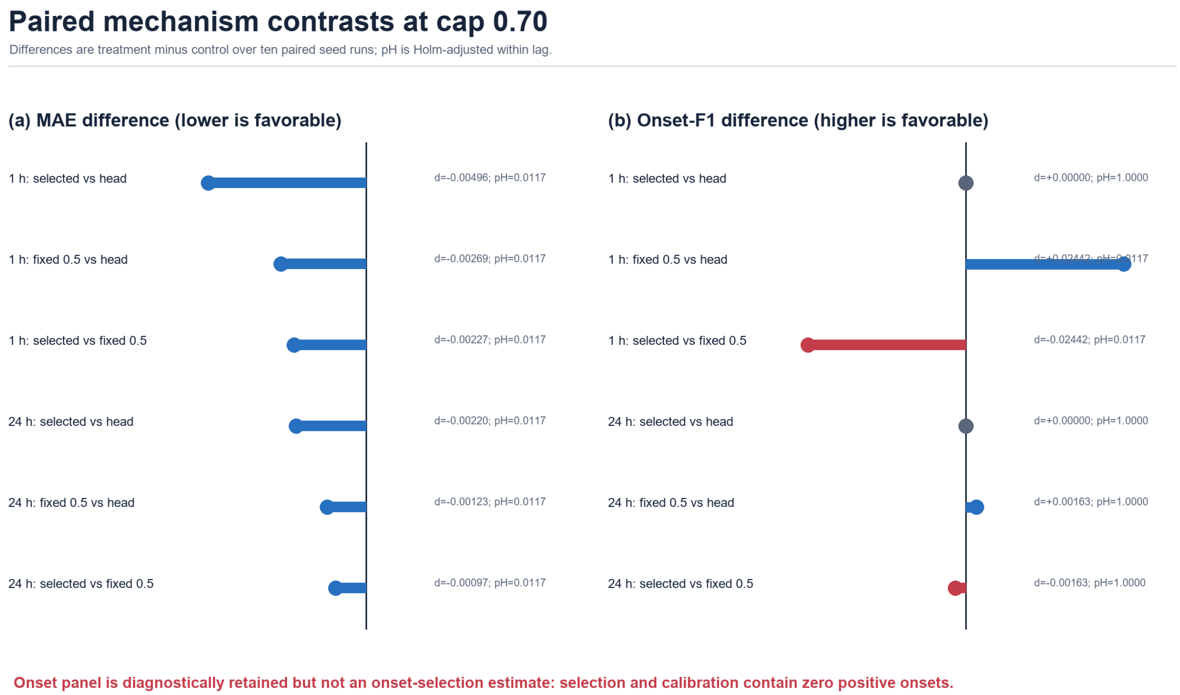

FIGURE 4. **Mean paired treatment-minus-control differences over ten common seeds. All MAE contrasts favor the condition containing more retrieval. The onset panel is retained to show the metric-specific negative and null results, but it is not an onset-selection estimate because pre-test onset support is absent.**

TABLE 6. **Descriptive method-level cap sensitivity from** `fair_cap_selected_vs_persistence.csv`**.**

Cap	Lag	Selected GRU-LSR MAE	Persistence MAE	GRU-LSR minus Persistence	Descriptive lower-MAE condition
0.60	1 h	0.01326705	0.01336988	-0.00010283	GRU-LSR
0.60	24 h	0.04384275	0.04301798	+0.00082476	Persistence
0.70	1 h	0.00777391	0.00690794	+0.00086597	Persistence
0.70	24 h	0.02076857	0.02054651	+0.00022206	Persistence
0.80	1 h	0.00356056	0.00280581	+0.00075475	Persistence
0.80	24 h	0.00790535	0.00804469	-0.00013934	GRU-LSR

targeted retrieval procedure would choose or benefit from retrieval when positive pre-test examples exist.

This distinction matters for the horizon-specific findings. Fixed 0.5 blending improves onset F1 at 1 h under the fallback, whereas the 24 h comparison is null. Reporting only the first result would hide both the failed selection premise and the 24 h null. Conversely, treating the selected-versus-head ties as evidence of no effect would confuse inapplicability with a supported null. Additional years or a prospectively frozen temporal design with positive selection and calibration onsets are required before RQ2 can be estimated.

C. NAIVE-REFERENCE AND CAP DEPENDENCE

Persistence remains lower-MAE than MAE-selected GRU-LSR at both primary-cap lags. The paired component result therefore does not imply overall forecasting superiority. The descriptive cap scan further shows that the ordering crosses twice and otherwise favors Persistence. This instability is scientifically relevant because the cap changes the proxy's event density and scale on the same underlying series. It also precludes a general statement that retrieval dominates or is dominated across policy settings.

D. INFORMATION AND APPLICATION BOUNDARY

The analysis unit is a paired seed run, and all results condition on one RTS-GMLC system and the first 8760 rows of the hashed 8784-row source files. Repeated hourly targets provide the evaluation surface but not independent replication. The direct transform demonstrates that target-hour scenario rows can reconstruct the proxy exactly, while the missing issuance and vintage fields prevent an operational information gate from being audited. The benchmark supports retrospective mechanism comparison only. It supplies no evidence about day-ahead deployment, operator decisions, economic value, dispatch feasibility, or performance under revised forecast vintages.

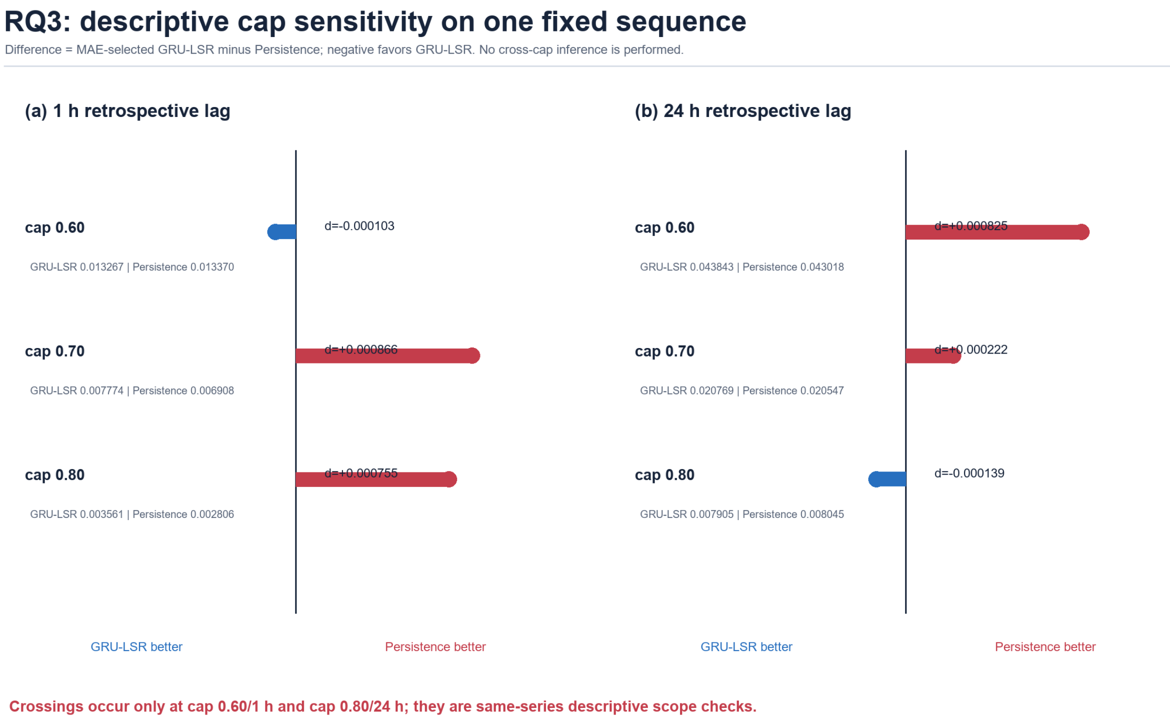

FIGURE 5. Selected GRU-LSR minus Persistence MAE at each cap and retrospective lag. Negative values favor GRU-LSR. The values are descriptive scope checks on one fixed 8760-row sequence; no cross-cap inferential claim is made.

VIII. LIMITATIONS

1. **Retrospective information only.** Source issue timestamps, as-of mappings, and data vintages are absent. The 1 h and 24 h tasks are delivery-row lag evaluations, not operational forecasts.
2. **Policy-derived proxy.** The target is defined by a fixed SNSP-type acceptance rule. It is not observed curtailment, an operator action, an AC-OPF solution, or a unit-commitment outcome.
3. **Zero pre-test onset support.** Every selection and calibration phase has zero positive onsets at both lags and all caps. Onset-targeted selection and threshold calibration use fallbacks and do not identify an onset effect.
4. **Single system and truncated source sequence.** The run uses the first 8760 of 8784 hashed source rows and ends on December 30, so it is not a complete calendar-year evaluation. No cross-year or cross-system accuracy result is available. Cap sensitivity reuses this sequence and is descriptive.
5. **Training-seed analysis unit and sign-flip assumption.** The ten pairs measure training randomness only. The sign-flip interpretation requires sign-exchangeability of paired differences under the no-effect null; the seeds are algorithmic initializations rather than randomized experimental assignments. The analysis does not quantify uncertainty across hours, onset blocks, years, systems, policies, or deployments, and no interval estimate over those units is available.
6. **Fair subset rather than full legacy roster.** The fair run does not rerun MLP, LSTM, DLinear, TCN, raw-feature kNN, SmallBank, encoder, or topology ablations. It cannot renew a full 14-method leaderboard or identify an overall fair-run winner.
7. **Limited mechanism attribution.** Raw-feature k-NN, randomized-encoder retrieval, alternative distances, and k sensitivity were not rerun under the fair gate. The paired result isolates the implemented retrieval prediction path from its head, but not learned geometry from generic neighbor averaging or smoothing.
8. **Point-prediction metrics.** Probabilistic calibration, utility-weighted errors, and event-block resampling are outside the executed protocol. The inherited onset classifier also does not encode the quiet-state prerequisite in its predicted-positive rule.
9. **No physical or user validation.** Nothing in the evidence establishes network feasibility, operator usefulness, deployment safety, or economic outcomes.

IX. CONCLUSION

This study provides a manifest-bound retrospective curtailment-risk benchmark and a fair matched evaluation of GRU learned-space retrieval on one RTS-GMLC system and a fixed 8760-row source sequence. The target is method-independent, and the fair run separates fit, head-first selection, calibration, test, and privileged visibility-control roles.

For continuous proxy error, MAE selection chooses retrieval only in all ten seeds. Relative to the matched GRU head, retrieval lowers MAE at both 1 h and 24 h retrospective lags (Holm-adjusted exact sign-flip $p = 0.01171875$ at each lag). The claim remains component-specific: Persistence is lower-MAE than selected GRU-LSR at the primary cap at both lags, and fixed 0.5 blending is worse than retrieval-only.

For onset detection, the fair data do not support target-matched selection because selection and calibration contain zero positive onsets at both lags. Selected GRU-LSR consequently equals the head ($p = 1$). A fixed 0.5 blend improves onset F1 at 1 h under the fallback, while the 24 h comparison is null; neither result validates onset-targeted retrieval. The direct transform's subunit onset F1 despite zero continuous error further exposes a metric-definition limitation. Across caps, GRU-LSR-versus-Persistence crossings are descriptive within the same fixed sequence.

The evidence therefore supports a paired MAE effect for the implemented retrieval path relative to its matched head, not a causal advantage of learned geometry, general forecasting superiority, an onset benefit, or operational readiness. Stronger conclusions require raw/randomized retrieval controls, issue-time/vintage-aware inputs, positive pre-test onset support, additional complete year and system units, and physical or user-centered validation.

X. ACKNOWLEDGMENT AND GENERATIVE-AI DISCLOSURE

[AUTHOR INPUT REQUIRED: confirm the final acknowledgment and venue-compliant generative-AI disclosure. The preparation record indicates AI-assisted drafting and code support, but the submitting authors must verify the tools, purposes, and responsibility statement.]

FUNDING

[AUTHOR INPUT REQUIRED: insert the funder and grant number, or explicitly state that the work received no external funding.]

XI. DATA AVAILABILITY AND REPRODUCIBILITY

RTS-GMLC is available from the GridMod repository (https://github.com/GridMod/RTS-GMLC) [6]. The completed fair-run package is stored under `experiments/p1_s3_fair_v1/` and contains the frozen configuration, executable script, manifest, 510-row result table, derived result tables, and policy-transform audit. `manuscript/figures/make_figures.py` validates the manifest-listed hashes before regenerating every paper figure and derived table. A separate rerun record documents reproduction of all scientific outputs. `SUPPLEMENTARY_METHODS_AND_AUDIT.md` retains environment, implementation-timing, version-history, checksum, and incident detail. No public release URL or archival DOI is claimed until the authors provide one.

XII. ETHICS DECLARATION

The executed study uses public synthetic power-system data and does not involve human participants, animals, or personal data. [AUTHOR INPUT REQUIRED: confirm the venue-specific ethics wording.]

AUTHOR CONTRIBUTIONS

[AUTHOR INPUT REQUIRED: provide a CRediT contribution statement. Contributions cannot be inferred from repository history.]

CONFLICTS OF INTEREST

[AUTHOR INPUT REQUIRED: confirm the conflicts-of-interest statement.]

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